

Program : Diploma in Cloud Computing and Big Data	
Course Code : 4271	Course Title: Big Data Concepts
Semester : 4	Credits: 4
Course Category: Engineering Science	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To familiarize with Big Data and its sources
- To understand the basics of data preprocessing and data visualization
- To familiarize Big Data tools and techniques.

Course Prerequisites:

Topic	Course Code	Course name	Semester
Understand the concept of Database systems		Database System Concepts	3
Understand the concept of Cloud Computing		Introduction to Cloud Computing	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the concepts of Big Data	16	Understanding
CO2	Explain the basic concepts in Big Data Analytics and NoSQL	14	Understanding
CO3	Describe Map Reduce programming in Hadoop Environment	16	Understanding
CO4	Outline the basic concepts of Hive and Pig	12	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						1
CO2	2						1
CO3	2	2					1
CO4	2						1

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the concepts of Big Data		
M1.01	Classify Digital Data	2	Understanding
M1.02	Explain Big data concepts	2	Understanding
M1.03	Outline the Characteristics of Big Data	2	Understanding
M1.04	Contrast Big Data and Data Warehouse	1	Understanding
M1.05	Describe Data Mining Concepts	3	Understanding
M1.06	Discuss Preprocessing techniques	3	Understanding
M1.07	Describe Data Visualization and List some tools used for Visualization	3	Understanding

Contents:

Classification of Digital Data: Structured, Unstructured and Semi Structured

Big Data: Characteristics of Data, Evolution of Big Data, Definition of Big Data, Challenges with Big Data, Need of big data

Characteristics of Big Data- Volume, Variety, Velocity, and other characteristics.

Applications, Big data sources, Traditional Business Intelligence (BI) versus Big Data

Data Warehouse: Definition, Coexistence of Big Data and Data Warehouse.

Data Mining- Introduction, Data Mining Goals, Types of data, Steps to discover knowledge from data(Stages of the Data Mining Process), Data Mining Techniques, Knowledge Representation Methods

Data Preprocessing -Need of data preprocessing, data quality, data preprocessing methods- data cleaning, data integration, data reduction, data transformation

Data Visualization: Definition, Need for Visualization, List some tools used for Visualization

CO2	Explain the basic concepts in Big Data Analytics and NoSQL		
M2.01	Outline the importance and challenges of Big Data Analytics	3	Understanding
M2.02	Define Data Science	1	Remembering
M2.03	Describe Responsibilities of a Data Scientist	1	Understanding
M2.04	Relate Big Data Environment terminologies	4	Understanding
M2.05	Explain the concepts of NoSQL	4	Understanding
M2.06	Summarize the concepts NewSQL	1	Understanding
	Series Test – I	1	Understanding

Contents:

Big Data Analytics: Definition, Classification of Analytics, Top Challenges Facing Big Data, Importance of Big Data Analytics, Technologies that meet challenges posed by Big Data.

Data Science: Definition, Responsibilities of a Data Scientist

Terminologies used in Big Data Environment: In-memory Analytics, In-Database Processing, Symmetric Multi-Processor Systems, Massively Parallel Processing, Difference Between Parallel and Distributed Systems, Shared-Nothing Architecture, CAP Theorem

NoSQL: Definition, Features of NoSQL, Types of NoSQL Databases, Advantages of NoSQL, Drawbacks of NoSQL, Use of NoSQL in Industry, NoSQL Vendors , SQL versus NoSQL

NewSQL: Characteristics of NewSQL, Comparison of SQL, NoSQL, and NewSQL

CO3	Describe Map Reduce programming in Hadoop Environment		
M3.01	Explain the fundamentals of Hadoop	4	Understanding
M3.02	Differentiate between RDBMS and HDFS	1	Understanding
M3.03	List the components of Hadoop Ecosystem	3	Understanding
M3.04	Illustrate the concept of HDFS and MapReduce	2	Understanding
M3.05	Summarize the different function in MapReduce	6	Understanding

Contents:

Hadoop- Features of Hadoop, Key Advantages of Hadoop, Versions, Hadoop Distributions, Hadoop versus SQL, Integrated Hadoop Systems Offered by Leading Market Vendors, Cloud-Based Hadoop Solutions

Advantages of Hadoop, RDBMS versus Hadoop , Distributed Computing Challenges, Overview of Hadoop Ecosystems and Hadoop Core Components, HDFS, Processing Data with Hadoop

MapReduce Programming: Introduction, Mapper, Reducer, Combiner, Partitioner, Searching, Sorting, Compression.

CO4	Outline the basic concepts of Hive and Pig		
M4.01	Explain the basic concepts of Hive	6	Understanding

M4.02	Describe the basic concepts of Pig	6	Understanding
	Series Test – II	1	
Contents: INTRODUCTION TO HIVE AND PIG Hive: Introduction – Architecture - Data Types - File Formats - Hive Query Language Statements – Partitions – Bucketing – Views - Sub- Query – Joins – Aggregations - Group by and Having - RCFile Implementation - Hive User Defined Function - Serialization and Deserialization (SERDE). Pig: Introduction - Anatomy – Features – Use Case for Pig - Pig Latin Overview - Pig Primitive Data Types - Running Pig - Execution Modes of Pig - HDFS Commands - Relational Operators - Eval Function - Complex Data Types - Piggy Bank - User-Defined Functions - Parameter Substitution - Diagnostic Operator - Word Count Example using Pig - Pig at Yahoo! - Pig Versus Hive			

Text / Reference

T/R	Book Title/Author
T1	Seema Acharya, Subhasini Chellappan, "Big Data Analytics", Wiley, 2015
R1	Jiawei Han,Champaign Micheline Kamber ,Jian Pei Simon Fraser "Data Mining Concepts and Techniques Third Edition"
R2	Frank J Ohlhorst, —Big Data and Analytics: Turning Big Data into Big Money, Wiley and SAS Business Series, 2012.
R3	Tom White, — Hadoop: The Definitive Guide Third Edition, O'reily Media, 2012
R4	EMC Education Services, Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data. John Wiley & Sons, 2015. Publishers – 2012.
R5	Michael Minelli, Michele Chambers, Ambiga Dhiraj, "Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends", John Wiley & Sons, 2013.

Online Resources

Sl.No	Website Link
1	https://hadoop.apache.org/
2	https://aws.amazon.com/emr/details/hadoop/what-is-hadoop/
3	https://en.wikipedia.org/wiki/Big_data
4	https://hive.apache.org/
5	https://pig.apache.org/
6	https://www.tutorialspoint.com/hadoop/index.htm